

Sequences and Series — Complete Notes

Grade 12

Mathematics

CAPS

What you'll learn

- The difference between a sequence (a list) and a series (a sum).
- How to use the arithmetic formulas for the n th term and the sum.
- How to use the geometric formulas for the n th term and the sum.
- When a geometric series converges, and how to find the sum to infinity.
- How to read and evaluate sigma notation.

The big picture

A **sequence** is an ordered list of numbers that follows a rule. For example, 3, 7, 11, 15, ... is a sequence because each term is 4 more than the one before it.

A **series** is what you get when you **add** the terms of a sequence together: $3 + 7 + 11 + 15 + \dots$

In CAPS you study two families — **arithmetic** (constant difference) and **geometric** (constant ratio). Learn to spot which one you are dealing with in the first ten seconds of a question, because every formula depends on that choice.

Key notation

- a = the first term.
- d = the common difference (arithmetic).
- r = the common ratio (geometric).
- n = the position number of a term.
- T_n = the n th term.
- S_n = the sum of the first n terms.

Arithmetic sequences

An arithmetic sequence has a **constant difference** d between consecutive terms: $T_2 - T_1 = T_3 - T_2 = d$.

n th term: $T_n = a + (n - 1)d$

Sum of the first n terms: $S_n = n/2 [2a + (n - 1)d]$

If the last term l is known you can use the shorter form: $S_n = n/2 (a + l)$.

Arithmetic: 3 7 11 15 19
 \ / \ / \ / \ /
 +4 +4 +4 +4 (constant difference $d = 4$)

Geometric sequences

A geometric sequence has a **constant ratio** r between consecutive terms: $T_2/T_1 = T_3/T_2 = r$.

nth term: $T_n = a \cdot r^{(n-1)}$

Sum of the first n terms (for $r \neq 1$): $S_n = a(r^n - 1)/(r - 1)$

The alternative form $S_n = a(1 - r^n)/(1 - r)$ is equivalent; use whichever gives fewer negative signs.

Geometric: 2 6 18 54
 \ / \ / \ /
 x3 x3 x3 (constant ratio $r = 3$)

Sum to infinity

If the ratio r satisfies $-1 < r < 1$ (that is, $|r| < 1$), the terms shrink towards zero and the series **converges** to a finite total:

$$S_\infty = a / (1 - r)$$

If $|r| \geq 1$ the terms do not shrink, so the sum to infinity does **not** exist. For $r = 1$ every term is the same and the total grows without limit; for $r = -1$ the partial sums keep flipping between two values.

The condition $|r| < 1$ is written as $-1 < r < 1$ and must always be stated when you use the formula.

Sigma notation

The symbol Σ (sigma) means "add up". In

Σ (from $k = 1$ to n) of T_k

k is a counter that runs from 1 to n , and each T_k is added. For example:

$$\Sigma \text{ (from } k = 1 \text{ to } 4) \text{ of } (2k + 1) = 3 + 5 + 7 + 9 = 24$$

Recognise the structure: if the terms form an arithmetic or geometric pattern, replace the sigma with the correct sum formula.

Worked example 1 — arithmetic

The sequence 7, 11, 15, ... is arithmetic. Find the 20th term and the sum of the first 20 terms.

Step 1 — identify a and d. $a = 7$ and $d = 11 - 7 = 4$.

Step 2 — nth term. $T_{20} = 7 + (20 - 1)(4) = 7 + 76 = 83$.

Step 3 — sum. $S_{20} = 20/2 [2(7) + (20 - 1)(4)] = 10 [14 + 76] = 10 \times 90 = 900$.

So $T_{20} = 83$ and $S_{20} = 900$.

Worked example 2 — geometric

In a geometric sequence the second term is 6 and the fifth term is 162. Find r , a and the first six terms.

Step 1 — write what you know. $T_2 = a \cdot r = 6$ and $T_5 = a \cdot r^4 = 162$.

Step 2 — divide to eliminate a. $(a \cdot r^4) / (a \cdot r) = 162 / 6 \rightarrow r^3 = 27 \rightarrow r = 3$.

Step 3 — find a. $a \cdot 3 = 6 \rightarrow a = 2$.

The first six terms are 2, 6, 18, 54, 162, 486.

Worked example 3 — sum to infinity and sigma

Evaluate Σ (from $k = 1$ to 8) of $3 \times 2^{(k-1)}$.

Step 1 — recognise the pattern. The terms are 3, 6, 12, 24, ... This is geometric with $a = 3$ and $r = 2$.

Step 2 — apply the geometric sum formula. $S_8 = 3(2^8 - 1) / (2 - 1) = 3(256 - 1) = 3 \times 255 = 765$.

Step 3 — check convergence separately. Because $r = 2$ and $|2| \geq 1$, this series does **not** have a sum to infinity.

How to recognise the sequence type

Before choosing a formula, test the pattern two ways:

Test	Arithmetic	Geometric
Subtract consecutive terms	same number every time	not constant
Divide consecutive terms	not constant	same number every time

Quick check: for 5, 10, 20, 40 the differences are 5, 10 and 20 (not constant), but the ratios are 2, 2 and 2 (constant), so it is geometric.

Some sequences are neither. A quadratic sequence such as 1, 4, 9, 16 has a constant **second** difference, and CAPS sometimes asks for the general term of such a pattern. The two families in this chapter, however, are arithmetic and geometric.

What convergence really means

A geometric series only settles down to a fixed total when each term is smaller than the one before it. If $r = 1/2$, the terms 8, 4, 2, 1, $1/2$, ... keep halving, and the running total creeps ever closer to 16 without ever passing it. That fixed ceiling is the sum to infinity.

When $-1 < r < 0$ the terms alternate in sign but still shrink, so the series also converges. When $|r| \geq 1$ the terms do not shrink towards zero, so the running total never settles, and the sum to infinity does not exist.

Reading word problems

Many CAPS questions hide a sequence inside a story — savings plans, salary increases, population growth or stacks of bricks. The workflow is always the same:

1. List the first three or four numbers.
2. Decide whether they are arithmetic or geometric.
3. Translate the question into "find T_n ", "find S_n " or "find S_∞ ".
4. State a and d , or a and r , then substitute into the matching formula.

If a question asks "how many terms are needed", you are solving for n , which usually leads to a quadratic and a rejected negative answer.

Common mistakes

- **Mixing up T_n and S_n .** T_n gives one term; S_n gives a total. Read the question carefully.

- **Using the wrong sign.** In $S_n = a(r^n - 1)/(r - 1)$ a negative r can flip signs — work slowly and show every line.
- **Forgetting the convergence condition.** You may only use $S_\infty = a/(1 - r)$ when $-1 < r < 1$.
- **Assuming every sequence is arithmetic.** Test the ratio as well as the difference before you choose a formula.
- **Off-by-one errors.** The first term has $n = 1$, so T_n uses $(n - 1)$, never n .
- **Not showing the formula.** CAPS awards method marks; write the formula before you substitute.

Quick summary / cheat sheet

Concept	Arithmetic	Geometric
Definition	constant difference d	constant ratio r
n th term	$T_n = a + (n-1)d$	$T_n = a \cdot r^{(n-1)}$
Sum of n terms	$S_n = n/2[2a + (n-1)d]$	$S_n = a(r^n - 1)/(r - 1)$
Sum to infinity	never (unless $d = 0$)	$S_\infty = a/(1 - r)$, $-1 < r < 1$

Memory hook: **arithmetic adds, geometric multiplies.**

Practice questions

1. Find the 15th term of the arithmetic sequence 5, 9, 13, ...
2. Find the sum of the first 12 terms of the arithmetic series with $a = 3$ and $d = 5$.
3. For the geometric sequence with $a = 4$ and $r = -2$, find T_6 and S_6 .
4. Find the sum to infinity of $8 + 4 + 2 + \dots$
5. A geometric sequence has $T_3 = 12$ and $T_6 = 96$. Find a and r .
6. How many terms of the series $3 + 7 + 11 + \dots$ are needed for the sum to equal 465?

Answers

1. $a = 5$, $d = 4$: $T_{15} = 5 + 14(4) = \mathbf{61}$.
2. $S_{12} = 12/2[2(3) + 11(5)] = 6(6 + 55) = 6 \times 61 = \mathbf{366}$.
3. $T_6 = 4(-2)^5 = 4(-32) = \mathbf{-128}$. $S_6 = 4[(-2)^6 - 1] / (-2 - 1) = 4(64 - 1)/(-3) = 252/(-3) = \mathbf{-84}$.
4. $a = 8$, $r = 1/2$ and $-1 < r < 1$, so $S_\infty = 8 / (1 - 1/2) = 8 / (1/2) = \mathbf{16}$.
5. $r^3 = 96/12 = 8 \rightarrow \mathbf{r = 2}$; then $a \cdot r^2 = 12 \rightarrow 4a = 12 \rightarrow \mathbf{a = 3}$.
6. $S_n = n/2[2(3) + (n-1)(4)] = n/2(4n + 2) = 2n^2 + n$. Set $2n^2 + n = 465 \rightarrow 2n^2 + n - 465 = 0 \rightarrow (2n + 31)(n - 15) = 0 \rightarrow \mathbf{n = 15}$ (the negative root is rejected).

Official EduRank Team notes. Always cross-check with your textbook and teacher.